Lesson 2 Inclined Plane

Objective: You will be able to:

-Define inclined plane

-Make inclined plane

-Use inclined plane in daily life.

-Calculate mechanical advantage.

1-Inclined plane

Do you know why the engineer build the curly road up a mountain?



An inclined plane is a flat supporting surface tilted at an angle, with one end higher than the other, used as an aid for raising or lowering a load.

Conservation of work on inclined Plane

Work done by load: $W_R = F_R \times h$

Work done by effort: $W_E = F_E \times L$



If no friction on inclined plane, work is conserved (Conservation of work):

$$W_R = W_E \Leftrightarrow F_R \times h = F_E \times L$$

This relation shows that the longer length of plane the less effort needed.

2.Mechanical advantage

The output load force is just the gravitational force of the object on the plane, its weight $\boldsymbol{F_R}$. The input force is the force $\boldsymbol{F_E}$ exerted on the object.

$$MA = \frac{Load}{Effort} = \frac{F_R}{F_E} = \frac{L}{h}$$

The MA of an ideal inclined plane without friction is sometimes called *ideal mechanical advantage*.

Where L is the length of ramp, h the height of ramp

Efficiency is defined as useful work divided by input work.

$$Rd = \frac{W_0}{W_{in}} = \frac{F_R \times d_R}{F_E \times d_E}$$

Practice Problems

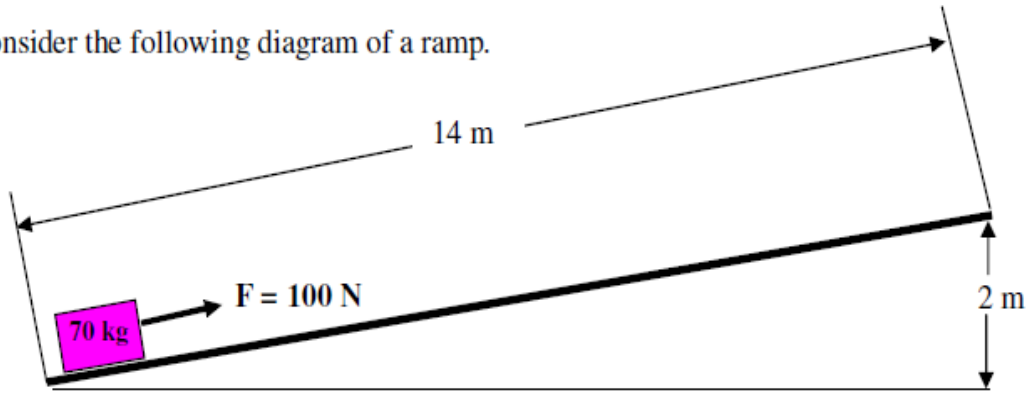
1-It takes 100 N to pull an object up an inclined plane. The gravitational force on the object is 600 N.

What is the load force in this case? …….N

What is the effort force in this case? ……..N

Calculate the Mechanical Advantage (MA)……

2. A 5-meter ramp lifts objects to a height of 0.75 meters. What is the mechanical advantage of the ramp?
3. A 10-meter long ramp has a mechanical advantage of 5. What is the height of the ramp?
4. A ramp with a mechanical advantage of 8 lifts objects to a height of 1.5meters. How long is the ramp?
5. A child makes a ramp to push his toy dump truck up to his sandbox. If he uses 5 Newton of force to push the 12-newton truck up the ramp, what is the mechanical advantage of his ramp?
6. A ramp with a mechanical advantage of 6 is used to move a 36-newton load. What input force is needed to push the load up the ramp?
7. Gina wheels her wheelchair up a ramp using a force of 80 Newton. If the ramp has a mechanical advantage of 7, what is the output force (in Newton)?
8. Consider the following diagram of a ramp.



a. Calculate the **force of gravity** on the 70kg object _____N

b. Calculate the **work** which would be done to raise the 70 kg mass straight up a vertical distance of 2 meters. (That is, without using the ramp.)

c. Calculate the **work** done when the object is moved 14 meters up the ramp using a force of 100 N.

d. What is the advantage of using a ramp like this to move an object up instead of lifting it straight up?

9-It takes 130 N to push a 52 kg object up an inclined plane. Calculate the Mechanical Advantage of the inclined plane, Show all your work.

10-A box weighing 100 Newton is pushed up an inclined plane that is 5 meters long. It takes a force of 75 Newton to push it to the top, which has a height of 3 meters. What is the efficiency of the machines?

11- Calculate the Mechanical Advantage of the inclined plane.

